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11 September 1987
SUPERSEDING
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MILITARY SPECIFICATION

MILLING MACHINES, VERTICAL SPINDLE, KNEE TYPE, SIZE NUMBERS 2, 3, 4, AND 5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sizes 2, 3, 4 and 5, vertical, knee type, electric motor driven milling machines with equipment and accessories specified herein.

1.2 Classification. The knee type milling machines shall be of the following sizes and classes. The size and class to be furnished shall be as specified herein.

Size 2 - 28 inches longitudinal table travel
Size 3 - 34 inches longitudinal table travel

Class 1 - 7-1/2 horsepower spindle drive
Class 2 - 10 horsepower spindle drive

Size 4 - 42 inches longitudinal table travel
Size 5 - 48 inches longitudinal table travel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B5.18 - Spindle Noses and Tool Shanks for Milling Machines.

ANSI B5.45 - Milling Machines.

ANSI B11.8 - Drilling, Milling, and Boring Machines, Safety Requirements for the Construction, Care, and Use.

ANSI/ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3308.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

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(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The milling machine shall be new and one of the manufacturer's current models. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The machine shall be designed in accordance with ANSI B5.45.

3.2.1 Measurement system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used on graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

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3.2.1.2 Digital readout (DRO) system. Unless otherwise specified (see 6.2.1), a DRO system shall be provided on the all of travel. The DRO system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0005 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0002 inch, measured over the full travel on each axis. Special requirements for the DRO system components shall be as specified (see 6.2.1). The DRO shall indicate axis location in both the US and SI units. The DRO system shall be provided with electronic and manual calibration to the machine on which it is installed to eliminate the effects of distortion error.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination of oil, dust, chips, coolants, and airborne particles.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual digits not less than 0.560 inch high. The console shall be capable of displaying at least six digits simultaneously. The display console shall also provide keyboard entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, absolute zero recall, and calculator add and subtract. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operation of the milling machine. The cable shall be securely fastened to the structure of the milling machine in accordance with the requirements of the contract.

3.2.2 Reclaimed materials. The milling machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use, performance, or design life. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the milling machine.

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3.2.3 Energy efficiency. The milling machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the milling machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the unit, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 and ANSI B11.8 that are applicable to the milling machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Mercury restriction. The milling machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the milling machine.

3.2.7 Environmental protection. The milling machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. An automatic lubrication system shall be provided for all ways and axes screws. A low lube warning light for this system shall be installed in plain view of the operator. The lube reservoir and pump for this system shall be self-contained, easily accessible, adjustable, and have adequate capacity to sustain the unit for a minimum of 24 hours of continuous operation without replenishing the reservoir.

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances. Foreign sourced replacement parts shall be interchangeable with domestically supplied components or be logistically supportable and readily available from a domestic source. Replacement parts are those that are normally subject to wear such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall

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also be considered replacement parts. All replacement parts shall be completely interchangeable without need for any modifications.

3.3 Construction. The milling machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solders or pastes shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swagged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts used on the milling machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.3.8 Gears. All gears and pinions of spindle and axis drive train shall be designed and manufactured of suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

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3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The milling machine shall consist of a column and base, knee, saddle, worktable, spindle, electric motor drive, and such other components as are required to perform as specified herein.

3.4.1 Column and base. The column and base shall be either one-piece or two piece construction with access doors or removable covers for servicing the drive motor, main drive clutch, and the adjusting mechanism. The walls shall be ribbed for rigidity. The column shall have machined ways of the square or dovetail design for vertical travel of the knee. Unless otherwise specified (see 6.2.1), all ways shall be hardened and precision ground for smooth accurate travel. Means shall be provided for securing the machine to the floor.

3.4.2 Knee. The knee shall be of the box-type construction with cross or transverse ribbing. The knee shall be fitted to the column ways with adjustable gibs. The ways shall have wipers or other means for preventing dust and chips from entering the ways. Means shall be provided to clamp the knee in any position within its travel on the column. The knee elevating screw shall have either square or acme threads and shall run in a bath of oil. The knee adjusting crank or handwheel shall positively and automatically release when power feed or rapid traverse is engaged. The knee shall have ways to support the saddle in its cross movement. Unless otherwise specified (see 6.2.1), the ways shall be either replaceable hardened and ground inserts or hardened as part of the knee and precision ground.

3.4.3 Saddle. Unless otherwise specified (see 6.2.1), all way surfaces of the saddle shall be hardened and precision ground. The ways shall be equipped with adjustable gibs and guides, as necessary, to maintain alignment in cross and longitudinal planes. Wipers or other means shall be provided to prevent dust and chips from entering ways. Means shall be provided for clamping the saddle to the knee in any position within its travel on the knee. The cross feed crank or handwheel shall positively and automatically disengage when power feed or rapid traverse is engaged.

3.4.4 Table. The table shall have not less than three equally spaced, manufacturer's standard size T-slots conforming dimensionally with ANSI B5.1. The table working surface shall be the manufacturer's standard size, but shall be not less than the size shown in table I, applicable to the size and class machine procured. The top of the table shall have a finished surface. The front face of the table shall be a finished surface having a slot or slots running the full length of the table for mounting adjustable table stops. The table shall have coolant drain channels with a chip screen and provisions for return of coolant to the reservoir. Means shall be provided for clamping the table to the saddle at any point in its range of travel. The table feed screw

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shall have a mechanism to control backlash in longitudinal table movement during climb milling operations. When specified (see 6.2.1), when backlash control is set and engaged for one table movement direction, the backlash eliminator shall automatically disengage during rapid traverse or conventional power feed in the opposite direction and shall automatically re-engage when original table movement is achieved for climb milling.

3.4.5 Spindle. The spindle shall be heat treated, hardened, and ground to a precision finish. It shall be supported by not less than two precision tapered roller bearings or two pairs of precision angular contact bearings that can be adjusted for axial alignment and not less than one other precision bearing of either the ball or roller type for radial support of the spindle. The spindle shall be equipped with a fly wheel or fly wheel integral with the bull gear, and shall be operable in both directions. The spindle shall have a precision ground nose in accordance with ANSI B5.18 and table I. An automatic braking mechanism shall stop the spindle when a lever or push button is placed in the off position or when the power to the spindle motor is turned off. Spindle speed changes shall be accomplished by movement of not more than three levers, a knob, or a push button with a direct reading indicator to show selected speed or operation.

3.4.6 Feed mechanism. The feed changes shall be accomplished by means of a single rotating lever, knob, or push button with a direct reading open faced dial to indicate the feed rate engaged. The feed control levers shall be directional and shall allow independent reversing of each feed. The feed mechanism shall be equipped with safety overload protection devices. Power feed shall automatically stop when the spindle is stopped.

3.4.7 Power rapid traverse. Power rapid traverse, independent of feed changes, shall be provided for the table in both directions of longitudinal, vertical, and cross travel. Power rapid traverse rates shall be in accordance with table I.

3.4.8 Stops. Adjustable and positive stops shall be provided for the knee, the table, and the saddle.

3.4.9 Clamping levers. Levers or screws shall be provided for clamping the knee to the column, the saddle to the knee, and the table to the saddle.

3.4.10 Feed screws and bearings. The feed screws for all axes shall be hardened and ground and shall be mounted in precision type ball or roller bearings.

3.4.11 Controls. All operating controls and all manual adjustments shall be grouped in a location convenient to the operator except as otherwise specified herein. All front handwheels and cranks shall be safety interlocked or shall automatically and positively disengage during power rapid traverse and power table feeds. When specified (see 6.2.1), the machine shall be provided with controls for starting and stopping the spindle, and initiating any table movement from the front and rear of the machine.

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3.4.12 Automatic cycle table control. When specified (see 6.2.1), milling machines shall have automatic cycle table control to provide for automatic work cycles by controlling power feed, rapid traverse, automatic reverse, and stop.

3.4.12.1 Automatic knee retraction. When specified (see 6.2.1), automatic knee retraction to lower the table during power rapid traverse shall be provided on the machines having automatic cycle table control.

3.4.13 Coolant. A coolant system shall be furnished. The system shall be a flood or mist type as specified (see 6.2.1). The coolant system shall include a sump or reservoir, a power driven pump, motor, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a strainer or baffle which is easily removable for cleaning. A coolant delivery nozzle and valve shall be provided to permit the operator to direct and control the amount of coolant flow. The shut-off control shall be independent of flow control.

3.4.14 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.15 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The milling machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.15.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-hertz (Hz). Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall be in accordance with ANSI/NEMA MG 1.

3.4.15.2 Solid-state components. Solid-state design shall be used throughout for electrical components.

3.4.15.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by low-voltage transformer(s) having secondary nominal voltage(s) of not greater than 120 volts.

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3.4.15.4 Hour meter. When specified (see 6.2.1), the milling machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.4.15.5 Electrical load indicator. Unless otherwise specified (see 6.2.1), the machine shall have a meter to indicate the electrical load on the spindle drive motor. The meter shall be current actuated. The dial shall indicate either amperes or percent of rated full-load current, and shall be clearly marked to indicate the rated reading for continuous operation of the machine. The accuracy of the meter shall be within 2 percent of full-rated load. The meter shall be clearly readable from the operator's normal position.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the milling machine shall conform to the requirements of table I.

3.6 Performance. Each machine shall be capable of satisfactory performance when tested in accordance with 4.7.1 through 4.7.4.

3.6.1 Additional performance requirements. Performance features required in addition to or in lieu of the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

1 set of wrenches normally furnished for the machine by the manufacturer
1 work light

3.9 Optional equipment. Optional equipment specified in the following subparagraphs shall be furnished with each machine as specified (see 6.2.1).

3.9.1 Universal dividing head. Unless otherwise specified (see 6.2.1), the universal dividing head shall be the standard commercial model and size normally furnished by the manufacturer for the size and class machine required. The dividing head shall be designed to index work and rotate work in conjunction with longitudinal movement of the table. It shall be operable with the front end of the spindle inclined at any angle from 0 degrees horizontal plane to 90 degrees vertical plane. The standard index plates shall be those normally furnished with the standard model dividing head. When specified (see 6.2.1), sufficient additional index plates shall be furnished

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to provide for dividing a single spindle rotation into any integral number of equal divisions from 2 to 50 when using the plain indexing method. When specified (see 6.2.1), a complete driving mechanism with enclosed gears shall be provided to drive the dividing head by power from the table feed screw. The spindle nose taper of the dividing head shall be the manufacturer's standard. The dividing head shall be complete with centers, steady rest, and 3-jaw universal chuck with reversible jaws. The tailstock shall be complete with bolts, nuts, washers and centers. An index table and a table of leads for cutting spirals shall be provided. When specified (see 6.2.1), a 6-jaw universal chuck for the universal dividing head shall be furnished.

3.9.2 Rotary table. When required, a rotary table with periphery in degrees shall be furnished complete with a handwheel, T-slots, T-bolts, nuts, washers, and locking clamp for rotary motion. The rotary table shall have a diameter as specified (see 6.2.1). When specified (see 6.2.1), a power drive unit and two trip dogs shall be furnished. When specified (see 6.2.1), an indexing attachment with the required number of indexing plates shall be furnished.

3.9.3 Swivel vise. When required, a swivel vise with the swivel base graduated in degrees shall be furnished complete with T-bolts, nuts, washers, crank, and a jaw opening as specified (see 6.2.1). The vise shall consist of a plain vise mounted on a detachable swivel block so as to be convertible to a plain vise.

3.9.4 Quill type spindle. When specified (see 6.2.1), a manual or power quill type spindle shall be furnished in lieu of a sliding head. The quill head shall be capable of swiveling either side of vertical position a minimum of 45 degrees.

3.9.5 Hardened and ground ballscrews. When specified (see 6.2.1), each sliding member shall be equipped with hardened and ground ball screws and recirculating ball-nut assemblies.

3.9.6 Independent axes drives. When specified (see 6.2.1), the machine shall be equipped with separate direct-current (DC) servo motors and solid-state drives to provide independent feeds for each axis. If independent axis drives are specified, the machine shall be equipped with hardened and ground ball screws and recirculating ball-nut assemblies nuts (see 3.9.5).

3.10 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

3.11 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following:

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Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant nameplate shall be securely attached to each milling machine and contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number of Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the milling machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

3.14 Training. Training required by the procuring activity shall be provided as specified in the contract.

3.15 Warranty. Warranty requirements shall be as specified in the contract by the procuring activity. The warranty period shall commence upon completion of installation and final acceptance testing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own Continental United States facility or any other Continental United States facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to

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perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of temperature: 55°F to 85°F; humidity: 10 percent to 90 percent relative humidity; and atmospheric pressure: 15.0 to 13.0 psi.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or any test shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.1 through 4.7.4, and the inspection in 4.8. Failure of the item to pass the examination, the test, or the inspection shall be cause for rejection.

4.6 Examination. The milling machine shall be visually examined to ensure compliance with all requirements of this specification. This examination shall also verify that the machine electrical system and components are in accordance with paragraphs 3.4.15 and 3.4.15.1 and all applicable standards cited in section 2.

4.7 Tests.

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4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment tolerance test. Each machine shall be tested to determine that it complies with the alignment tolerance requirements of 3.7.

4.7.3 Performance tests.

4.7.3.1 Horsepower, milling accuracy, and tool chatter tests. Each machine shall be tested by machining a workpiece of SAE 4140 steel. The workpiece shall be not less than 16 inches long, 6 inches wide, and 3 inches thick. The workpiece may be drilled to facilitate clamping. One of the two surfaces having the largest area shall be ground flat and mounted in contact with the table. Roughing cuts shall be made to mill metal off the horizontal and one vertical surface of the workpiece. The roughing cuts shall be made at sufficient depth and feed rate to load the spindle drive motor to not less than maximum continuous duty rated electrical input. The type of cutting tool, depth of cut, spindle speed, and feed rate shall be determined by the manufacturer, provided the required loading is achieved. There shall be no evidence of chatter, stalling, or permanent distortion during or after the test. Finishing cuts shall be made on the horizontal and vertical surfaces. The finished surfaces shall meet the following accuracies:

4.7.3.2 Squareness. The two milled sides shall be tested for squareness. The sides shall be square with each other within 0.001 inch per 4 inches.

4.7.3.3 Flatness. The two milled sides shall be tested for flatness. The sides shall be flat within 0.001 inch per foot.

4.7.3.4 Parallelism. The finished horizontal surface shall be tested for parallelism. The top surface shall be parallel to the bottom surface within 0.001 inch per foot.

4.7.4 Climb milling test. Each machine shall be tested in accordance with the manufacturer's standard climb milling test, provided that the test proves the capability of the machine to meet all of the requirements for which it was designed.

4.7.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

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5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The milling machines covered by this specification are to be used for conventional milling operations on a wide range of parts.

6.2 Ordering Data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

c. First article, if required (see 3.1).

d. If the milling machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Digital readout system, specify axes readout(s) required, if different, (see 3.2.1.2).

g. Specify digital readout requirements (see 3.2.1.2).

h. Additional safety and health requirements, if required (see 3.2.4).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

k. Column and base ways, if different, specify requirement (see 3.4.1).

l. Knee ways, if different, specify requirement (see 3.4.2).

m. Saddle ways, if different, specify requirements (see 3.4.3).

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n. Automatic backlash elimination device, if required, specify requirement (see 3.4.4).

o. Duplicate front and rear controls, if required (see 3.4.11).

p. Automatic cycle table control, if required (see 3.4.12).

q. Automatic knee retraction, if required (see 3.4.12.1).

r. Specify flood or mist type coolant system (see 3.4.13).

s. Electrical system, if different, specify requirement (see 3.4.15).

t. Type motor enclosure, if different (see 3.4.15.1).

u. Hour meter, if required, specify requirement (see 3.4.15.4).

v. Electrical load indicator, if not required (see 3.4.15.5).

w. Size and capacity, if different (see 3.5).

x. Additional performance features, if required (see 3.6.1).

y. Standard equipment, if different, specify requirement (see 3.8).

z. Optional equipment, as required (see 3.9).

aa. Universal dividing head, if other than standard commercial model (see 3.9.1).

bb. Additional indexing plates for dividing head, if required (see 3.9.1).

cc. Power drive unit for dividing head, if required (see 3.9.1).

dd. Six-jaw universal chuck for the dividing head, if required (see 3.9.1).

ee. Diameter of rotary table, if required (see 3.9.2).

ff. Power drive unit for rotary table, if required (see 3.9.2).

gg. Indexing attachment for rotary table, if required (see 3.9.2).

hh. Specify jaw opening size of swivel vise, if required (see 3.9.3).

ii. Quill type swiveling spindle in lieu of sliding type, if required; specify manual or power feed (see 3.9.4).

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jj. Hardened and ground ball screw and nut assemblies, if required (see 3.9.5).

kk. Independent axes drives, if required (see 3.9.6).

ll. Specify applicable repair parts, if required (see 3.10).

mm. Lubrication chart or plate, if different (see 3.11.1).

nn. Nameplate, if different (see 3.11.2).

oo. Inspection conditions, if different (see 4.3).

pp. First article inspection, if different (see 4.4).

qq. Quality conformance inspection, if different (see 4.5).

rr. Preservation, packaging, and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Machine tools
Milling machine, knee
Spindle, vertical

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Sizes and capacities.

Characteristics	Size 2	Size 3		Size 4	Size 5
		Class 1	Class 2		
Table:					
Travel, inches:					
Longitudinal	28	34	34	42	48
Cross	9	12	12	16	16
Vertical, knee	14	14	14	17	17
Vertical, head	6	6	6	6	6
Quill travel (option see 3.9.4)	3	3	3	5	5
Working surface	49 x 10	55 x 12-3/4	63 x 13-3/8	71 x 15-3/4	78-3/4 to 16
Number of feeds	16	18	16	16	16
Optional quill feeds	4	4	4	4	4
Feed rate range, IPM:					
Longitudinal	1/2 to 31	1/2 to 31	1/2 to 31	1/2 to 31	1/2 to 31
Cross	1/2 to 31	1/2 to 31	1/2 to 31	1/2 to 31	1/2 to 31
Vertical	9/64 to 9	9/64 to 9	9/64 to 9	9/64 to 9	9/64 to 9
Feed motor horsepower	2	2	4	4	4
Power rapid traverse, IPM:					
Longitudinal	118	118	118	118	118
Cross	118	118	118	118	118
Vertical	35	35	35	35	35
Spindle:					
Drive motor hrosepower	5	7-1/2	10	15	20
ANS taper	50	50	50	50	50
Number of speeds	12	12	16	16	16
Speed range, RPM	28 to 1400	28 to 1400	28 to 1400	28 to 1400	28 to 1400

NOTE 1: Machine size shall not be less than the stated requirements of the size specified in 6.2.1, nor shall it exceed the size requirements specified in 6.2.1 by more than 10 percent.

NOTE 2: An infinitely variable feed rate is acceptable provided it meets the requirements and ranges shown in the above table.

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TABLE II. Alignment accuracies.

Description	Allowable deviation, not more than
Spindle:	
Axial runout	0.0003 inch TIR
Radial runout	0.0004 inch TIR
Taper runout:	
Measured 1 inch from spindle	0.0004 inch TIR
Measured 8 inches from spindle	0.0006 inch TIR
Spindle square with table surface	0.001 inch per foot
Transverse rise and fall of table	0.001 inch per foot
Longitudinal rise and fall of table	0.001 inch per foot
Knee travel parallel to spindle	0.001 inch per foot
Head travel square with table surface	0.0004 inch per 4 inches
Center T-slot parallel to table travel	0.001 inch per foot
Movement of knee when clamp is locked	0.001 inch

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - YD
Air Force - 84
DLA - GS

User activities:
Navy - MC

Preparing activity:
DLA-IP

(Project 3417-0177)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-80051D	Milling Machines, Vertical Spindle, Knee Type, Size Numbers 2, 3, 4, and 5

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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